BRAC University (Department of Computer Science and Engineering)
CSE 221 (Algorithms) for Spring 2026 Semester
(NO EXTRA PAGE IS ALLOWED)
Quiz 1
SET - A

Student ID: **Section:** **Full Marks: 20**
Name: **Duration: 25 minutes**

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1. Write down the time complexity of the following code snippet in **Big-O** notation. Here, n is an input variable and k is a constant. **(5 marks)**

```
for (int i=1; i*i<=n; i++)
{
   int sum = 0;
   for (int j=n; j>=1; j=j/k)
   {
      for (int m=1; m<=n; m+=3){
         if ( m%2==0) break;
         sum = sum + 10;
      }
   }
}
```

2. For attending the quiz, you have earned **2 marks**.

3. Write down the rules/steps/assumptions for converting a growth function to asymptotic time analysis. **(3 marks)**

4. You are given an array which denotes a unimodal function with ***a plateau***. The sequence is strictly decreasing, then the minimum element may repeat several times (this region is called plateau), then strictly increasing. A sample input can be like the following array:

Index	1	2	3	4	5	6	7	8	9	10	11	12
Value	10	8	7	6	4	4	4	4	5	8	9	13

(i) Draw the function and identify the plateau region. **(2 marks)**

(ii) Design a condition or property which will divide the array into two parts (First all T then F, or all F then T). **(3 marks)**

(iii) Write down the code for binary search for determining the rightmost minimum element in the plateau region. **(5 marks)**

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Quiz 2

Student ID: **Full Marks: 20** **Section:**
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1. You are given **three sorted arrays A, B, and C** of lengths n1, n2, and n3, respectively. Each array is sorted in non-increasing order. Write the pseudocode/code for a function ***3-Merge (A, B, C)*** that merges these three arrays into a **single sorted array D** of length n = n1 + n2 + n3. Your algorithm must run in **O(n) time. (10 marks)**

2. For attending the quiz, you have earned **2 marks.**
3. Show the simulation of the Partitioning step for the following array (**Pivot is 13**). **(8 marks)**

Index	1	2	3	4	5	6	7	8
Value	6	10	**13**	5	8	30	2	11

Bonus (3 marks)

Now consider a variant of MergeSort called **3-MergeSort**, where instead of dividing the array into two halves, the input array of size *n* is divided into three equal parts recursively. After sorting the three subarrays, the algorithm uses the 3-Merge(A, B, C) function to combine them. Is the time complexity of 3-MergeSort better than MergeSort? Give an explanation.

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Quiz 3

Student ID: **Full Marks: 20** **Section: 20**
Name: **Duration: 15 minutes**

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1. Write one difference and one similarity between connected components and strongly connected components in a graph. **(5 marks)**

2. For attending the quiz, you have earned **2 marks.**

3. Any graph always has a topological sorting. Do you agree or disagree with the statement? Explain your reasoning. **(3 marks)**

4. Draw a graph with 10 vertices where there will be 9 tree edges, 6 back edges, 3 cross edge and 2 forward edges. You need to show the labeling of the edges clearly. **(7 marks)**

5. How will you reconstruct a cycle after finding a back edge? **(3 marks)**

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Quiz 4

Student ID: **Full Marks: 20** **Section: 21**
Name: **Duration: 15 minutes**

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1. At a university, there are N students, numbered from 1 to N. Initially, each student belongs to their own separate club. Over time, some clubs decide to collaborate and merge into one larger club. You need to manage these club connections and answer some queries.

   Join (A, B): Student A's club and student B's club merge into one club. *If they are already in the same club, do nothing.*
   Same (A, B): Check whether student A and student B belong to the same club. Print 'Yes' if they are in the same club, 'No' otherwise.

   I. If N = 7, then write down the parent array. **(2 marks)**

   II. For the following sequence of operations, manually simulate the parent array from I after each operation using **Disjoint Set Union (DSU)** and also draw the trees accordingly: **(10 marks)**

   a. Join (4, 2)
   b. Join (2, 3)
   c. Same (1, 4)
   d. Join (1, 5)
   e. Join (7, 5)
   f. Same (4, 7)

2. For attending the quiz, you have earned **2 marks.**
3. A graph with negative edge weights will always contain a negative cycle. Do you agree with the statement or not? Provide examples or explain your reasoning. **(8 marks)**

Bonus: (2 marks)
Explain path compression with a proper example. You must show the necessary trees and parent array.